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11. The IUPAC name of the compound $CH_3-\underset{\underset{OH}{|}}{CH}-COOH$ is

a) 2 – Hydroxypropionic acid

b) 2 – Hydroxy Propanoic acid

c) Propan – 2– ol –1 – oic acid

d) 1 – Carboxyethanol.

12. The IUPAC name of $\begin{matrix} CH_3 \\ \diagdown \\ \quad CH \\ \diagup \\ Br \end{matrix} - \underset{\underset{CH_3}{|}}{CH} - COOH$ is

a) 2 – Bromo -3 – methyl butanoic acid

b) 2 - methyl - 3- bromobutanoic acid

c) 3 - Bromo - 2 - methylbutanoic acid

d) 3 - Bromo - 2, 3 - dimethyl propanoic acid.

13. The structure of isobutyl group in an organic compound is

a) $CH_3 - CH_2 - CH_2 - CH_2-$

b) $CH_3 - \underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}$

c) $CH_3 - \underset{\underset{CH_3}{|}}{CH} - CH_2 -$

d) $CH_3 - \underset{|}{CH} - CH_2 - CH_3$

14. The number of stereoisomers of 1, 2 – dihydroxy cyclopentane

a) 1 b)2 c) 3 d) 4

15. Which of the following is optically active?

a) 3 – Chloropentane

b) 2 Chloro propane

c) Meso – tartaric acid

d) Glucose

16. The isomer of ethanol is

a) acetaldehyde b) dimethylether

c) acetone d) methyl carbinol

17. How many cyclic and acyclic isomers are possible for the molecular formula C_3H_6O?

a) 4 b) 5 c) 9 d) 10

18. Which one of the following shows functional isomerism?

a) ethylene b) Propane

c) ethanol d) CH_2Cl_2

19. $\overset{\ominus}{CH_2}-\underset{\underset{O}{||}}{C}-CH_3$ and $CH_2 = \underset{\underset{O}{|}}{\overset{\ominus}{C}} - CH_3$ are

a) resonating structure b) tautomers

c) Optical isomers d) Conformers.

20. Nitrogen detection in an organic compound is carried out by Lassaigne's test. The blue colour formed is due to the formation of.

a) $Fe_3[Fe(CN)_6]_2$

b) $Fe_4[Fe(CN)_6]_3$

c) $Fe_4[Fe(CN)_6]_2$

d) $Fe_3\ [Fe(CN)_6]_3$

21. Lassaigne's test for the detection of nitrogen fails in

a) $H_2N - CO- NH.NH_2.HCl$

b) $NH_2 - NH_2.\ HCl$

c) $C_6H_5 - NH - NH_2.\ HCl$

d) $C_6H_5\ CONH_2$

22. Connect pair of compounds which give blue colouration / precipitate and white precipitate respectively, when their Lassaigne's test is separately done.

a) $NH_2\ NH_2\ HCl$ and $ClCH_2–CHO$

b) $NH_2\ CS\ NH_2$ and $CH_3 - CH_2Cl$

c) $NH_2\ CH_2\ COOH$ and $NH_2\ CONH_2$

d) $C_6H_5NH_2$ and $ClCH_2 - CHO$.

23. Sodium nitropruside reacts with sulphide ion to give a purple colour due to the formation of

a) $[Fe(CN)_5\ NO]^{3-}$